



Problem Programming Assignment 1

1 Written Portion

Answer the following questions and submit your typed responses in a PDF in your submission. You will be asked to write proofs for this section. [Here](#) is a good resource for writing proofs that mirrors what is expected out of written answers for this class.

Q1: Describe your heuristic. Include the equation/algorithm that you used.

A1: The heuristic uses the Manhattan distance $|x_1 - x_2| + |y_1 - y_2|$ divided by the maximum jump length in the board

Q2: Write a proof showing that the heuristic you chose is *admissible*. Recall that for a heuristic $h(s)$ and a true cost to the goal $g(s)$, h can be considered admissible if $h(s) \leq g(s)$ for all possible states.

A2:

Let $h(n)$ be the cost of the heuristic and $h(n')$ be the cost to the goal.

The cost = the number of moves regardless of jump length

Let a cell $n = (r, c)$ in the grid, and cell $g = (r_g, c_g)$ in the grid

Let L be the maximum jump length of cells in the grid. L will never be negative

A single move can only affect one row or column at a time, and by at most L :

$$MD(j, k) = |r - r_k| + |c - c_k|$$

You can only move at most L spaces

To get from cell n to cell g , the heuristic determines $h(s) = (|r - r_g| + |c - c_g|)/L$

Take any path from n to g , of cost s , and assume the length of the optimal path is $s = h'(n)$. $s = h'(n)$ is an integer as you cannot make a fractional number of moves

Since s is the cost of the optimal path and L is the maximum jump length of any cell, then $\sum_{i=1}^s L$ is maximum jump distance for the optimal path

Assume for sake of contradiction that $\sum_{i=1}^s L < MD(n, g)$. This means the optimal cost s multiplied by the maximum jump length is less than the lowest amount of spaces between the start and end cells, thus s would not be minimum cost because you cannot reach the destination, thus it follows that:

$$MD(n, g) \leq \sum_{i=1}^s L$$

$$MD(n, g) \leq sL$$

$$s \geq MD(n, g)/L$$

As $s = h'(n)$ and $MD(n, g)/L = h(n)$

Then $h'(n) \geq h(n)$

Q3: Write a proof showing that the heuristic you chose is *consistent*. Recall that for a heuristic $h(s)$ and a cost function $c(n \rightarrow n')$, that h is consistent if $h(n) \leq c(n \rightarrow n') + h(n')$ for any two neighbors n and n' .

A3:

Let n and n' be cells in the grid such that n' is a neighboring cell of n

Let g be the goal cell in the grid

The distance between n and $n' \leq L$ as L is the maximum jump length

$$c(n \rightarrow n') = 1$$

$$|MD(n, g) - MD(n', g)| \leq L$$

Case $MD(n, g) > MD(n', g)$

$$MD(n, g)/L - MD(n', g)/L \leq 1$$

$$MD(n, g)/L \leq 1 + MD(n', g)/L$$

$$h(n) \leq c(n \rightarrow n') + h(n')$$

Case $MD(n, g) \leq MD(n', g)$

$$MD(n, g)/L \leq MD(n', g)/L$$

$$MD(n, g)/L \leq MD(n', g)/L < 1 + MD(n', g)/L$$

$$h(n) \leq c(n \rightarrow n') + h(n')$$